

Commissioned Review: Branched-Chain Amino Acid Catabolism (KEGG ppu00280) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** `kegg:ppu00280` — "Valine, leucine and isoleucine degradation" **Module area:** amino acid metabolism **Candidate genes reviewed:** 35 local metadata entries **Purpose:** support manual module satisfiability and gene-annotation curation

1. Executive summary

Branched-chain amino acid (BCAA) catabolism is **present and satisfiable** in *P. putida* KT2440. The pathway is organized as a shared "upper" hub feeding three substrate-specific "lower" arms, and every dedicated biochemical step maps to a credible KT2440 gene. There is **no true gap** in the core route from Val/Leu/Ile to the terminal C3/C2 products (propionyl-CoA and acetyl-CoA).

The upper hub is the best-characterized part of the pathway in this exact organism: branched-chain aminotransferase (**ilvE**, PP_3511) generates the three branched 2-oxo acids, which are then oxidatively decarboxylated by the **bkd**-encoded branched-chain 2-oxo acid dehydrogenase complex (BCKDH: **bkdAA/bkdAB/bkdB/lpdV**, PP_4401–PP_4404). The *bkd* operon of *P. putida* has been cloned, sequenced, and shown to be strictly required for growth on BCAAs, controlled by the Lrp-family activator BkdR and by Crc-mediated catabolite repression — this is direct, target-strain evidence. Downstream, the leucine arm is complete and dedicated (**ivd/liuA** PP_4064 → **mccA/mccB** PP_4067/PP_4065 → **liuC** PP_4066 → HMG-CoA lyase PP_3394 or **mvaB** PP_3540; equivalent to KEGG module M00036), the valine arm is carried by an **mmsAB**-like operon (**mmsB** PP_4666 + **mmsA-II** PP_4667), and the isoleucine arm is carried largely by promiscuous fatty-acid β -oxidation enzymes.

The dominant curation issues are therefore **not** missing steps but **over-inclusion and paralog ambiguity**. Roughly 15 of the 35 `ppu00280` members are generic β -oxidation enzymes (thiolases, acyl-CoA dehydrogenases, enoyl-CoA hydratases, fadA/fadB) co-mapped into the

pathway purely through shared EC numbers/KOs, not because they are dedicated BCAA enzymes. Several genuine steps have 2–5 paralogs (E3 dihydrolipoamide dehydrogenase, HMG-CoA lyase, methylmalonate-semialdehyde dehydrogenase) that require explicit "which copy" decisions. Finally, the module boundary should be drawn at propionyl-CoA/acetyl-CoA: propionyl-CoA is disposed via the **2-methylcitrate cycle** (separate KEGG maps), not the B12 methylmalonyl-CoA-mutase module, and the **atu**/acyclic-terpene entry present in some pseudomonads is **absent** in KT2440.

2. Target-organism pathway definition

2.1 Exact process included

ppu00280 covers the **oxidative catabolism of the three branched-chain amino acids** — L-valine, L-leucine, and L-isoleucine — from the free amino acids to the point where their carbon skeletons enter central metabolism as **acetyl-CoA, acetoacetate, and propionyl-CoA**. Mechanistically this comprises:

1. **Transamination** of Val/Leu/Ile to their 2-oxo (branched-chain keto) acids (BCKAs);
2. **Oxidative decarboxylation** of the three BCKAs by a shared multienzyme BCKDH complex, producing the branched acyl-CoA thioesters (isobutyryl-CoA, isovaleryl-CoA, 2-methylbutyryl-CoA);
3. **Substrate-specific "lower" arms** that process each branched acyl-CoA by a β -oxidation-like sequence (dehydrogenation, carboxylation/hydration, cleavage) to the terminal products.

2.2 Neighboring pathways to keep separate

- **Fatty acid degradation (map00071 / ppu00071) and general β -oxidation.** Many BCAA lower-arm enzymes belong to the same superfamilies (acyl-CoA dehydrogenases, enoyl-CoA hydratases, 3-hydroxyacyl-CoA dehydrogenases, thiolases) as generic fatty-acid β -oxidation enzymes and share EC numbers. These generic enzymes are co-mapped into **ppu00280** but should be treated as **shared/peripheral**, not dedicated.
- **Propanoate metabolism (map00640) and the 2-methylcitrate cycle.** Propionyl-CoA generated by the valine and isoleucine arms is **not** part of BCAA degradation proper; it is assimilated downstream.

- **Synthesis and degradation of ketone bodies / terpenoid degradation.** The leucine terminal steps overlap the HMG-CoA node (ppu00907/ketogenesis), and in some pseudomonads the Liu pathway is shared with acyclic terpene (citronellol/geraniol) catabolism via the **atu** cluster — which KT2440 lacks.
- **BCAA biosynthesis (ppu00290, valine/leucine/isoleucine biosynthesis).** Note that **ilvE** (branched-chain aminotransferase) and **ldh** (leucine dehydrogenase) sit at the biosynthesis/degradation interface and appear in ppu00290-labeled buckets in the metadata.

2.3 Alternate names / database definitions

- KEGG map: "**Valine, leucine and isoleucine degradation**" (map00280); organism-specific `ppu00280`.
- KEGG module: **M00036 "Leucine degradation, leucine => acetoacetate + acetyl-CoA"** captures the dedicated leucine arm.
- Functional cluster names in *Pseudomonas*: **bkd** operon (BCKDH), **liu** = leucine/isovalerate utilization (liuRABCDE in *P. aeruginosa*), **mms** = methylmalonate-semialdehyde pathway (valine lower arm), **atu** = acyclic terpene utilization (absent in KT2440).

3. Expected step model

The pathway in KT2440 can be drawn as a **Y-shaped hub-and-arms** topology:



Terminal products / module boundary: acetyl-CoA and acetoacetate (leucine), propionyl-CoA (valine and isoleucine). The module should **terminate at propionyl-CoA/acetyl-CoA**.

4. Candidate genes and evidence

4.1 High-confidence, dedicated BCAA genes

Gene	Locus	Step	EC / KO	Evidence type	Curation note
ilvE	PP_3511	Transamination Val/Leu/Ile → BCKAs	2.6.1.42 / K00826	Homology + strong enzymology in related bacteria	Sits at biosynthesis/degradation interface (metadata bucket ppu00290); a single BCAT typically serves all three substrates
bkdAA	PP_4401	BCKDH E1 α	1.2.4.4 / K00166	Direct, target strain	Core hub; part of cloned/sequenced <i>bkd</i> operon
bkdAB	PP_4402	BCKDH E1 β	1.2.4.4 / K00167	Direct, target strain	Core hub
bkdB	PP_4403	BCKDH E2 (dihydrolipoamide acyltransferase)	2.3.1.- / K09699	Direct, target strain	Core hub
lpdV	PP_4404	BCKDH E3 (dihydrolipoamide DH, dedicated)	1.8.1.4 / K00382	Direct, target strain	Dedicated E3 co-encoded in <i>bkd</i> operon (see paralog note 5.2)
ivd/ liuA	PP_4064	Isovaleryl-CoA DH (leucine arm)	1.3.8.4 / K00253	Homology; strong <i>P. aeruginosa</i> enzymology	Leucine arm entry; KM for isovaleryl-CoA 2.3 μ M in PAO1 LiuA
mccB/ liuB	PP_4065	3-Methylcrotonyl-CoA carboxylase β	6.4.1.4 / K01969	Homology; <i>P. aeruginosa liu</i> cluster	Biotin-dependent carboxylase
mccA/ liuD	PP_4067	3-Methylcrotonyl-CoA carboxylase α (biotin carboxylase)	K01968	Homology; <i>P. aeruginosa liu</i> cluster	Forms MCase with mccB
liuC	PP_4066	3-Methylglutaconyl-CoA hydratase	4.2.1.18 / K13766	Homology; <i>P. aeruginosa liu</i> cluster	Leucine arm

Gene	Locus	Step	EC / KO	Evidence type	Curation note
mmsB	PP_4666	3-Hydroxyisobutyrate DH (valine arm)	1.1.1.31 / K00020	Homology; <i>P. aeruginosa</i> mms enzymology	Adjacent to mmsA-II → mms operon
mmsA-II	PP_4667	Methylmalonate-semialdehyde DH (valine arm)	1.2.1.27 / K00140	Homology + genomic-context QC (this review)	Correct valine-arm copy (see 5.2)

4.2 Leucine terminal step (HMG-CoA lyase) — displaced paralogs

Gene	Locus	EC / KO	Note
PP_3394	PP_3394	4.1.3.4 / K01640	3-hydroxy-3-methylglutaryl-CoA lyase; leucine terminal step candidate
mvaB	PP_3540	4.1.3.4 / K01640	Second HMG-CoA lyase paralog

The KT2440 *liu* cluster is only PP_4064–4067 and **lacks a co-located liuE** (HMG-CoA lyase), unlike the *P. aeruginosa* liuRABCDE cluster. The terminal leucine step is therefore **displaced** to PP_3394 or mvaB/PP_3540. Curation should mark the HMG-CoA lyase step **covered but non-adjacent**, and flag the paralog choice for review.

4.3 Alternative entry: leucine dehydrogenase

Gene	Locus	EC / KO	Note
ldh	PP_4617	1.4.1.9 / K00263	Leucine dehydrogenase — oxidative deamination of leucine as an alternative to ilvE transamination

This provides a redundant entry to 2-oxo-isocaproate for leucine; it is a legitimate alternative route, not a required step.

4.4 Generic / peripheral β -oxidation enzymes (likely over-mapped)

These are co-mapped into `ppu00280` through shared EC numbers/KOs but are **generic fatty-acid β -oxidation** or tangential enzymes, not dedicated BCAA steps:

- **Thiolases (K00626 / 2.3.1.9 / 2.3.1.16)**, $\times 5+$: PP_0582, PP_2215, PP_3355, **bktB**/PP_3754, **yqeF**/PP_4636, plus *fadA* copies PP_2051, PP_2137.
- **Acyl-CoA dehydrogenases (K00249 / 1.3.8.8)**, $\times 3$: PP_2437, PP_2793, PP_3725; plus **acd**/PP_2216.
- **Enoyl-CoA hydratases (K01692 / 4.2.1.17)**, $\times 2+$: PP_2217, **paaF**/PP_3284; and multifunctional **fadB**/PP_2136.
- **Tangential nodes**: PP_0596 (β -alanine/omega-amino acid aminotransferase, K00822), **aacs**/PP_3071 (acetoacetyl-CoA synthetase, EC 6.2.1.16), **atoA/atoB** (PP_3122/PP_3123, 3-oxoacid CoA-transferase, K01028/K01029), **mmsA-I**/PP_0597 (second methylmalonate-semialdehyde DH paralog, genomically isolated).

These should generally be marked **shared/peripheral** for the module rather than counted as satisfying dedicated steps — though one or more acyl-CoA DH / enoyl-CoA hydratase / thiolase does carry the **isoleucine lower arm** (which has no dedicated cluster in KT2440).

5. Gaps, ambiguities, and likely over-annotations

5.1 Over-inclusion (the primary curation problem)

Approximately **15 of 35** `ppu00280` members are generic β -oxidation enzymes pulled in by EC/KO sharing (Finding F004). Left uncured, they inflate apparent pathway membership and obscure which genes are truly dedicated. The dedicated leucine module M00036 is fully covered by a small set of specific genes; the bulk of the remaining membership is peripheral.

5.2 Paralog ambiguity (requires explicit "which-copy" decisions)

- **E3 dihydrolipoamide dehydrogenase (K00382 / EC 1.8.1.4)** — **3 paralogs**: **lpdV**/PP_4404 (dedicated, *bkd*-encoded), *lpdG*/PP_4187, and *lpd*/PP_5366 (housekeeping, shared with pyruvate and 2-oxoglutarate dehydrogenase). **Assign lpdV/PP_4404** as the BCKDH E3.
- **Methylmalonate-semialdehyde DH (K00140 / EC 1.2.1.27)** — **2 paralogs**: *mmsA-I*/PP_0597 vs *mmsA-II*/PP_4667. **Genomic-context QC in this review (F007)** shows PP_4667 is directly

adjacent to mmsB/PP_4666 (3-hydroxyisobutyrate DH) with a LysR-family regulator at PP_4668, forming an mms-like valine-catabolic operon, whereas PP_0597 is genomically isolated (neighbors: LysR regulator PP_0595, paraquat-inducible protein PP_0598). **Assign mmsA-II/PP_4667** to the valine arm.

- **HMG-CoA lyase (K01640 / EC 4.1.3.4) — 2 paralogs:** PP_3394 vs mvaB/PP_3540. Both are candidates for the displaced leucine terminal step; choice is **unresolved** and should be flagged.
- **Entry step redundancy:** ilvE/PP_3511 (transamination) vs ldh/PP_4617 (oxidative deamination, leucine only).

5.3 True gaps

None for the dedicated core. The isoleucine lower arm has no dedicated gene cluster and is inferred to run on promiscuous β -oxidation enzymes — this is best marked **covered-by-shared-enzymes / candidate_uncertain** rather than a gap.

5.4 Boundary / not-expected-in-target-taxon

- **atu / acyclic terpene entry: absent.** KT2440 lacks the atu cluster and cannot grow on acyclic terpenes (citronellol/geraniol), although it does use leucine and isovalerate (F002). Any terpene-linked entry steps are **not_expected_in_target_taxon**.
- **Propionyl-CoA disposal is out of scope.** Propionyl-CoA is routed via the **2-methylcitrate cycle** (separate maps), not the B12 methylmalonyl-CoA-mutase module M00741 (F006).

6. Module and GO-curation recommendations

Module step	Recommended status	Rationale
Transamination (Val/Leu/Ile → BCKAs)	covered (ilvE/PP_3511; ldh/PP_4617 alt for Leu)	Homology strong; single BCAT convention
BCKDH oxidative decarboxylation	covered (bkdAA/AB/B + lpdV, PP_4401-4404)	Direct target-strain evidence
Leucine arm (ilvD→mcc→liuC→HMG-CoA lyase)	covered (M00036 complete)	Dedicated genes; terminal step displaced to PP_3394/mvaB
Valine arm (... →mmsB→mmsA)	covered (mmsB/PP_4666 + mmsA-II/PP_4667)	Operon structure confirmed by genomic context
Isoleucine lower arm	candidate_uncertain	Carried by promiscuous β -oxidation enzymes; no dedicated cluster
HMG-CoA lyase paralog choice	candidate_uncertain	PP_3394 vs mvaB/PP_3540 unresolved
Generic β -oxidation members (≈ 15)	peripheral / module_needs_revision	Over-mapped by shared EC/KO; should not count as dedicated steps
atu/terpene entry	not_expected_in_target_taxon	Cluster absent in KT2440
Propionyl-CoA → central metabolism	out_of_scope (2-methylcitrate cycle)	Belongs to propanoate maps, not ppu00280

Module boundary recommendation: define `ppu00280` for KT2440 to **terminate at propionyl-CoA + acetyl-CoA/acetoacetate**. Explicitly exclude the propionyl-CoA assimilation steps and the methylmalonyl-CoA-mutase module.

GO / annotation actions: demote the ~ 15 generic β -oxidation members from "dedicated BCAA degradation" to shared β -oxidation membership; annotate lpdV, mmsA-II, and (tentatively) one HMG-CoA lyase as the dedicated copies; retain ilvE and ldh both as legitimate but redundant entry options.

7. Genes to promote to full fetch-gene review

1. **PP_4401–PP_4404 (bkdAA/bkdAB/bkdb/lpdV)** — core hub with direct target-strain evidence; anchor the module and confirm E3 assignment vs lpdG/lpd.
2. **PP_3394 vs PP_3540 (mvaB)** — resolve which HMG-CoA lyase paralog carries the displaced leucine terminal step.
3. **PP_4666/PP_4667 (mmsB/mmsA-II) vs PP_0597 (mmsA-I)** — confirm the valine-arm operon assignment and clarify the role (if any) of the isolated mmsA-I paralog.
4. **PP_3511 (ilvE) and PP_4617 (ldh)** — document the redundant entry routes and substrate ranges.
5. **Isoleucine lower-arm candidates** — identify which specific acyl-CoA DH / enoyl-CoA hydratase / 3-hydroxyacyl-CoA DH / thiolase paralogs carry 2-methylbutyryl-CoA processing (currently unassigned).

8. Mechanistic model / interpretation

The KT2440 BCAA catabolic network is best understood as a **funnel-then-fan** system. All three amino acids funnel through two shared steps — transamination (ilvE, with ldh as a leucine-specific backup) and the BkdR-regulated BCKDH complex — before fanning out into three chemically distinct lower arms. This architecture explains several curation observations:

1. **Why the hub is so well characterized and the arms less so.** The *bkd* operon is a single genetic and regulatory unit (BkdR activation, Crc repression, BCAA induction), making it a natural experimental target; the lower arms are distributed across the genome and overlap general β -oxidation. In *P. putida* the four structural genes were cloned and sequenced, and the positive regulator BkdR (an Lrp homolog, 36.5% identity to *E. coli* Lrp) is divergently transcribed and required for growth on BCAAs — *bkdR* mutants cannot use Val/Leu/Ile (PMID: 8320210). L-BCAAs and D-leucine are the physiological inducers (half-maximal L-Val $\approx$ 2.8 mM; PMID: 10217783), and the operon is under Crc-mediated catabolite repression (PMID: 10648542, PMID: 10648543). This is finding **F001**.

2. **Why over-mapping is the main annotation error.** Because the branched acyl-CoAs are processed by the same enzyme chemistry as fatty acids (dehydrogenation → hydration → oxidation → thiolytic cleavage), KEGG's EC/KO-based mapping sweeps many generic β -oxidation enzymes into `ppu00280`. The correct curation stance is a small **dedicated core** plus a large **shared periphery** (finding **F004/F005**).
3. **Why paralog resolution needs genomic context.** Where enzymology alone cannot distinguish paralogs (E3, mmsA, HMG-CoA lyase), **operon structure** is decisive: *lpdV* sits inside *bkd*; *mmsA-II* sits beside *mmsB* under a shared regulator; the isolated *mmsA-I* does not (finding **F007**).
4. **Why the boundary matters.** Drawing the module edge at propionyl-CoA/acetyl-CoA keeps `ppu00280` coherent and avoids double-counting the 2-methylcitrate cycle and (absent) B12-mutase route. In pseudomonads, propionyl-CoA from branched catabolism is channeled into central metabolism via the 2-methylcitrate cycle (PMID: 19798672), and the shared BCKDH readily acts on the isoleucine-derived 2-oxo acid (PMID: 19910413) — finding **F006**.

9. Evidence base and key references

Direct target-strain (*P. putida* KT2440 / *P. putida*) evidence

- **PMID: 8320210** — *The bkdR gene of Pseudomonas putida is required for expression of the bkd operon...* Establishes that the *bkd* operon and its Lrp-homolog activator BkdR are required for BCAA catabolism: "Chromosomal mutations affecting this gene, named *bkdR*, resulted in a loss of ability to use branched-chain amino acids as carbon and energy sources and failure to produce branched-chain keto acid dehydrogenase." (Supports F001.)
- **PMID: 10217783** — *In vitro transcriptional studies of the bkd operon...L-branched-chain amino acids and D-leucine are the inducers.* "Therefore, the L-branched-chain amino acids and D-leucine are the inducers of the *bkd* operon." Defines the physiological inducers of the shared BCKDH step. (Supports F001.)
- **PMID: 9068646**, **PMID: 8670279**, **PMID: 7836297** — BkdR transcriptional activation, DNA binding, and stoichiometry in *P. putida*; confirm the regulatory architecture of the hub.
- **PMID: 10648542**, **PMID: 10648543** — Crc-mediated catabolite repression of the *bkd* operon in *P. putida* (posttranscriptional control of BkdR). (Supports F001.)

- **PMID: 19910413** — Confirms the shared BCKDH ("the branched-chain OADHC (BCDHC) of *Pseudomonas putida* could be measured easily") acts on branched 2-oxo acids including the isoleucine-derived substrate. (Supports F006.)

Related-*Pseudomonas* evidence (transfer to KT2440: strong-to-moderate)

- **PMID: 16820476** — *Identification of genes...for catabolism of acyclic terpenes and leucine/isovalerate in P. aeruginosa*. Directly states KT2440 has candidate *liu* genes but not *atu*: "Analysis of the genome sequences of other pseudomonads (*P. putida* KT2440 and *P. fluorescens* Pf-5) revealed candidate genes for Liu proteins for both species and candidate genes for Atu proteins in *P. fluorescens*." And: "*P. fluorescens*, but not *P. putida*, could grow on acyclic terpenes (citronellol and citronellate), while both species were able to utilize leucine and isovalerate." (Supports F002.) *Transfer: strong — statement is explicitly about KT2440*.
- **PMID: 18625020** — Biochemical characterization of LiuA (isovaleryl-CoA DH, KM 2.3 μ M) and demonstration that *liu* genes are essential for methyl-branched compound catabolism in *P. aeruginosa*. Underpins the KT2440 leucine-arm assignment (ivd/PP_4064). *Transfer: moderate-to-strong (orthologous cluster)*.
- **PMID: 1339433** — *Characterization of the mmsAB operon of P. aeruginosa PAO...* Defines the valine lower arm: "The operon contains two structural genes involved in valine metabolism: *mmsA*, which encodes methylmalonate-semialdehyde dehydrogenase; and *mmsB*, which encodes 3-hydroxyisobutyrate dehydrogenase." Mutants "grow slowly on valine/isoleucine medium and exhibit reduced enzyme activity in cell-free extracts." (Supports F003; maps to PP_4666/PP_4667.) *Transfer: strong (orthologous operon confirmed by KT2440 genomic context in F007)*.

Broader / mechanistic context

- **PMID: 19798672** — *Pseudomonas* sp. M1 β -myrcene proteomics: "the beta-myrcene catabolic intermediate propionyl-CoA is channeled into the central metabolism via the 2-methylcitrate cycle." Supports the module boundary decision (propionyl-CoA disposal is out of scope). (Supports F006.)
- **PMID: 378964** — *E. coli* transaminase B (*ilvE*): a single branched-chain aminotransferase serves Val/Leu/Ile with no separate valine-specific activity — supports treating PP_3511 as the shared entry BCAT.
- **PMID: 35698914** — Crystal structure of leucine dehydrogenase (EC 1.4.1.9), supporting the *ldh*/PP_4617 alternative entry route.

- **PMID: 32997715** — *P. aeruginosa* FadE2 acyl-CoA dehydrogenase shows substrate preference for Val/Leu catabolic intermediates — illustrates how promiscuous acyl-CoA dehydrogenases participate in BCAA lower arms (relevant to the isoleucine-arm assignment problem).

10. Limitations and knowledge gaps

- **Arm-specific enzymology is inferred, not measured, in KT2440.** The leucine and valine lower-arm assignments rest on *P. aeruginosa* orthology plus KT2440 genomic context; individual KT2440 enzymes (LiuA/ivd, MccAB, LiuC, MmsAB) have not each been biochemically validated in the target strain.
- **Isoleucine lower arm is unresolved at the gene level.** No dedicated cluster; the specific promiscuous β -oxidation paralogs carrying 2-methylbutyryl-CoA processing are not individually assigned.
- **HMG-CoA lyase paralog choice (PP_3394 vs mvaB) is unresolved.**
- **This review is computational.** Findings derive from KEGG KO mapping, genomic-neighborhood inspection, and literature; no new wet-lab data were generated in the target strain.

11. Proposed follow-up experiments / actions

1. **Targeted deletions in KT2440** of ivd/PP_4064, mccAB, liuC, mmsB/mmsA-II, and each HMG-CoA lyase paralog, phenotyped on Val, Leu, and Ile as sole carbon sources — resolves arm assignments and the PP_3394/mvaB question directly in the target strain.
2. **^{13}C -labeling / metabolomics** with labeled Val/Leu/Ile to trace flux to propionyl-CoA vs acetyl-CoA/acetoacetate and confirm the 2-methylcitrate routing.
3. **Comparative genomics / synteny** across *P. putida* strains to confirm the mms and liu operon boundaries and the absence of liuE/atu.
4. **Curation actions:** (a) demote the ~15 generic β -oxidation members to shared periphery; (b) fix paralog assignments (lpdV, mmsA-II) using genomic context; (c) set the module terminus at propionyl-CoA/acetyl-CoA; (d) mark atu/terpene entry not_expected_in_target_taxon; (e) promote the genes in Section 7 to full `fetch-gene` review.

Prepared as a species-aware module/pathway/taxon review to support manual module satisfiability and gene-annotation curation for Pseudomonas putida KT2440.

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